

1. Cloud Concepts & Economics

- **Trade Fixed for Variable Expense:** Pay only for what you use instead of investing heavily upfront in data centers.
 - **Massive Economies of Scale:** AWS aggregate usage leads to lower pay-as-you-go prices for everyone.
 - **Stop Guessing Capacity:** Scale resources up and down dynamically based on real-time application load.
 - **Elasticity:** Automatically acquire resources during high load and release them when demand drops.
 - **Agility:** Drastically increases the speed to test, deploy, and launch new software projects in minutes.
 - **Global Footprint:** Deploy applications globally in minutes to achieve the lowest latency for worldwide users.
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2. AWS Infrastructure & Global Network

- **AWS Region:** A distinct geographic area hosting isolated clusters of data centers to meet local regulatory and latency needs.
 - **Availability Zone (AZ):** Discrete, redundant data centers within a Region. Deploying across multiple AZs creates a **highly available, fault-tolerant, and resilient architecture**.
 - **Edge Location:** Sites used by **Amazon CloudFront** to cache static website content closer to global users for low latency.
 - **Scale Hierarchy:** There are more Edge Locations than AZs, and more AZs than Regions.
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3. Core AWS Services

Compute

- **Amazon EC2:** Resizable virtual servers. Use **Auto Scaling groups** for the lowest operational cost under variable loads.
- **AWS Lambda:** Serverless compute that runs code for short, event-driven tasks with zero server management.

- **AWS Fargate:** Serverless compute engine explicitly designed to run **containers** without managing EC2 instances.
- **Amazon WorkSpaces:** Managed **virtual desktops** allowing remote employees to securely access environments from home devices.

Storage

- **Amazon S3:** Object storage with virtually unlimited capacity. Used for static websites and **WORM compliance** via **Glacier Vault Lock**.
- **Amazon Athena:** Direct, serverless **SQL syntax queries** of flat data objects sitting inside an S3 bucket.
- **Amazon EBS:** Block storage volumes attached to an individual EC2 instance. Backed up via operationally efficient **EBS Snapshots**.
- **Amazon EFS:** A fully managed, serverless **NFS (Network File System)** shared simultaneously across multiple compute resources.
- **AWS Storage Gateway (File Gateway):** Connects on-premises environments to S3 object storage via standard network protocols like **NFS**.

Databases

- **Amazon RDS:** Managed relational database service that handles provisioning, patching, updating, and Multi-AZ automatic failover.
- **Amazon DynamoDB:** Fully managed, single-digit millisecond **NoSQL database** built to scale to millions of continuous queries per second.
- **Amazon Neptune:** Fully managed **graph database** service optimized for highly connected datasets like real-time fraud pattern detection.

Networking

- **Amazon VPC:** A logically isolated, custom private network section within the AWS Cloud.
- **Internet Gateway:** A gateway attached to a VPC to enable inbound and outbound public internet access.
- **VPC Endpoint:** Connects a VPC securely to AWS services (like S3) **without using the public internet**.
- **AWS Direct Connect:** A dedicated, **private physical network connection** bypassing the public internet to deliver highly consistent throughput.

- **AWS Transit Gateway:** A centralized cloud router hub to interconnect multiple VPCs and on-premises networks.
 - **Amazon Route 53:** Highly available cloud Domain Name System (DNS) resolution service.
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4. Security, Identity, & Compliance

Firewalls & Traffic Filtering

- **Security Groups:** Instance-level virtual firewalls. They are **stateful** and control access to a single EC2 instance regardless of its subnet.
- **Network ACLs:** Subnet-level virtual firewalls. They are completely **stateless** filters for a VPC network.
- **AWS WAF:** Web Application Firewall used to filter inbound web traffic and explicitly block **SQL Injection attacks**.
- **AWS Shield Advanced:** Specialized protection against massive **DDoS attacks** providing real-time visibility and access to the DRT team.

Identity, Auditing, & Compliance

- **AWS IAM:** Manages fine-grained user access to AWS service APIs. **Always free of charge**. Follows the *Principle of Least Privilege*.
 - **AWS IAM Identity Center:** Central hub to manage single-sign-on access for remote employees to multiple AWS accounts and SaaS portals.
 - **AWS CloudTrail:** Records account API event history—logging exactly **who** initiated an action, **what** resource was affected, and **when** it happened.
 - **AWS Config:** Records, evaluates, and tracks **configuration changes** and compliance rules of AWS resources over time.
 - **AWS Artifact:** Your on-demand download portal for official compliance documents like **SOC** and **PCI** control reports.
 - **AWS Secrets Manager:** Securely stores, encrypts, and **automatically rotates** database credentials and API keys.
 - **AWS KMS:** Manages cryptographic keys to seamlessly **encrypt data at rest** across services.
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5. Billing, Support, & Frameworks

EC2 Optimization Strategy

- **On-Demand:** Best for short-term, unpredictable workloads that cannot be interrupted.
- **Reserved Instances / Savings Plans:** Best for predictable, steady-state, uninterruptible workloads running continuously for **1 to 3 years**.
- **Spot Instances:** Best for batch workloads that can handle interruptions to minimize costs.

Cost Management Tools

- **AWS Pricing Calculator:** Models architecture options to provide an **estimate of costs before migration**.
- **AWS Budgets:** Sets up notifications to alert you if spending patterns **exceed your planned budget limit**.
- **AWS Cost Explorer:** Visualizes historical spending trends, patterns, and projects future costs.

Well-Architected Pillars

- **Reliability:** Focuses on the ability to **recover automatically from failure** and testing recovery procedures.
- **Operational Excellence:** Continuous process improvement to deliver ongoing business value.
- **Performance Efficiency:** Efficient resource selection, emphasizing **using serverless architectures**.

6. Shared Responsibility Model

- **AWS (Security OF the Cloud):** Responsible for global facilities, physical data center security, hardware maintenance, edge locations, and the host virtualization layer.
 - **Customer (Security IN the Cloud):** Responsible for data encryption, updating/patching the guest operating systems on EC2, setting firewall/security group rules, and managing IAM user credentials.
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7. Support Plans

- **Basic Plan: Always free.** Gives access to 24/7 customer service and **AWS Developer Forums**.
- **Business Plan: Lowest cost** plan providing access to the **full set of AWS Trusted Advisor checks** and programmatic case support via API.
- **Enterprise Plan:** Includes **Infrastructure Event Management (IEM)** for critical software rollouts and assigns a dedicated **Technical Account Manager (TAM)**.

AWS Trusted Advisor: Core tool checking 5 pillars: **Cost Optimization, Security, Fault Tolerance, Performance, and Service Quotas**.

8. Missing Specialized AI & Analytics Services (From Section 9)

- **Amazon Polly:** An AI service that turns text into lifelike speech (Text-to-Speech) to create voice-enabled apps.
 - **Amazon Translate:** A neural machine learning service that delivers fast, high-quality, and affordable language translation.
 - **Amazon Textract:** Automatically extracts text, handwriting, and data from scanned legal or financial documents.
 - **Amazon Comprehend:** A Natural Language Processing (NLP) service that uses machine learning to find structural insights and sentiments in text.
 - **Amazon Redshift:** A fast, fully managed **cloud data warehouse** used to run complex analytics on massive datasets (Petabyte scale).
 - **Amazon EMR (Elastic MapReduce):** A big data cloud tool used to process vast amounts of data using open-source tools like Apache Spark and Hadoop.
 - **Amazon Kinesis:** Real-time data streaming service used to collect, process, and analyze video and data streams as it arrives.
 - **AWS Glue:** A serverless data integration service that makes it easy to discover, prepare, and combine data (ETL) for analytics.
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9. Developer Tools (The "AWS Code" Suite)

- **AWS CodeCommit:** A secure, highly scalable, managed source control service that hosts private Git repositories.

- **AWS CodeBuild:** Compiles source code, runs automated tests, and produces ready-to-deploy software packages.
- **AWS CodeDeploy:** Automates software deployments to any instance, including Amazon EC2 and on-premises servers.
- **AWS CodePipeline:** A continuous delivery service that automates your release pipelines for fast and reliable application updates.
- **AWS Elastic Beanstalk:** The easiest way to deploy web apps. You just upload your code (Java, .NET, PHP, Node.js, Python), and it automatically handles deployment, provisioning, and load balancing.